AGREEMENT OF THE K-GROUPS OF AN AFFINE SCHEME WITH THE K-GROUPS OF ITS RING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The zeroth K-group of a ring equals the zeroth K-group of its perfect derived category, by the More on Algebra universal property lemma. invoked at proof lines 10143-10144. Provides the module-level description of K_0(R).
- On an affine scheme perfect objects correspond to perfect complexes of modules and the derived categories compare, by the perfect affine and affine comparison lemmas. invoked at proof lines 10145-10147. Identifies D_perf(R) with D_perf(O_X), giving K_0(R) = K_0(X).
- Over a Noetherian ring the coherent sheaves on the affine scheme are the finite modules, by the cohomology coherent Noetherian lemma. invoked at proof lines 10152-10155. Gives K'_0(R) = K'_0(X) from the definitions.

Source reference. Original section 00JC: sources/stacks-chow/tags.txt, source lines 710–710. Cited at proof lines 10141-10142 for the definitions of K_0(R) and K'_0(R); the identifications are inside the counted span.

Source reference. Original lemma 0FCU: sources/stacks-chow/tags.txt, source lines 19194–19194. Cited at proof lines 10143-10144 for K_0(R) = K_0(D_perf(R)); the transport is inside the counted span.

Source reference. Original lemma 08EB: sources/stacks-perfect/pinned-original-source.tex, source lines 2636–2636. Cited at proof lines 10145-10146 to identify D_perf(R) with D_perf(O_X); the K_0 conclusion is inside the counted span.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 10145-10147 together with the perfect affine lemma; the K_0 conclusion is inside the counted span.

Source reference. Original lemma 01XZ: sources/stacks-chow/tags.txt, source lines 2538–2538. Cited at proof lines 10152-10153 for the K'_0 identification; the conclusion is inside the counted span.

Lemma 0.2. *Let $X = \mathrm{Spec}(R)$ be an affine scheme. Then $K_0(X) = K_0(R)$ and if R is Noetherian then $K'_0(X) = K'_0(R)$.*

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Proof. Recall that $K'_0(R)$ and $K_0(R)$ have been defined in Algebra, Section 00JC. By More on Algebra, Lemma 0FCU we have $K_0(R) = K_0(D_{perf}(R))$. By Lemmas 08EB and 06Z0 we have $D_{perf}(R) = D_{perf}(\mathcal{O}_X)$. This proves the equality $K_0(R) = K_0(X)$.

The equality $K'_0(R) = K'_0(X)$ holds because $Coh(\mathcal{O}_X)$ is equivalent to the category of finite R-modules by Cohomology of Schemes, Lemma 01XZ. Moreover it is clear that $K'_0(R)$ is the zeroth K-group of the category of finite R-modules from the definitions. $\square$